

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **Thamaraiselvi S (192511042)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Thamaraiselvi S**

Assessment Tasks

Scenario-Based Questions

1. A university plans to virtualize its computer laboratory so that multiple operating systems can be provided on demand. Recommend a virtualization strategy, identify the suitable hypervisor type, and justify the decision.
2. A hospital cloud is facing VM escape and unauthorized snapshot access. Identify the major security risks and propose mitigation controls.
3. A retail company wants to modernize its datacenter using Software Defined Datacenter principles. Explain how compute, storage, and network virtualization would be introduced in phases.
4. Two cloud providers want to interoperate through federation. Explain how trust, identity, and privacy issues should be handled in the federation design.
5. An enterprise observes poor performance in a virtualized environment after overcommitting CPU and memory to multiple VMs. Analyze the scenario and recommend tuning actions.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Scenario-Based Questions

1. A university plans to virtualize its computer laboratory so that multiple operating systems can be provided on demand. Recommend a virtualization strategy, identify the suitable hypervisor type, and justify the decision.

Answer:

The university should implement a **server virtualization strategy** using centralized virtual machines. A **Type-1 (Bare-Metal) Hypervisor** is the most suitable choice because it runs directly on the physical hardware without requiring a host operating system. Examples include VMware ESXi, Microsoft Hyper-V, and Xen.

With this approach, each student can access a virtual machine configured with different operating systems such as Windows, Ubuntu, or Fedora based on course requirements. The university can quickly create, clone, or restore VMs for different lab sessions, reducing hardware costs and simplifying maintenance.

A Type-1 hypervisor provides better performance, stronger security, and higher reliability than a Type-2 hypervisor. It also supports centralized management, live migration, snapshots, and resource allocation, making it ideal for educational environments where many students share the same infrastructure.

2. A hospital cloud is facing VM escape and unauthorized snapshot access. Identify the major security risks and propose mitigation controls.

Answer:

The hospital cloud faces two major security risks: **VM escape** and **unauthorized snapshot access**.

VM escape occurs when an attacker breaks out of a virtual machine and gains access to the host operating system or other virtual machines. This can expose sensitive patient records and compromise multiple healthcare applications.

Unauthorized snapshot access is another serious threat because VM snapshots may contain patient data, user credentials, encryption keys, and application states. If attackers obtain these snapshots, they can restore or copy confidential information.

To mitigate these risks, the hospital should:

- Keep the hypervisor updated with the latest security patches.

- Enable Secure Boot and Trusted Platform Module (TPM).
- Use Role-Based Access Control (RBAC) so only authorized administrators can create or access snapshots.
- Encrypt VM snapshots and backups.
- Enable Multi-Factor Authentication (MFA) for administrators.
- Continuously monitor virtual machines using Intrusion Detection and Security Information and Event Management (SIEM) tools.
- Perform regular vulnerability assessments and security audits.

These controls improve confidentiality, integrity, and availability of healthcare data while ensuring compliance with healthcare regulations.

3. A retail company wants to modernize its datacenter using Software Defined Datacenter (SDDC) principles. Explain how compute, storage, and network virtualization would be introduced in phases.

Answer:

The retail company should adopt a phased implementation strategy for the Software Defined Datacenter.

Phase 1 – Compute Virtualization

Physical servers are converted into virtual machines using a Type-1 hypervisor. Applications are migrated from physical servers to virtual machines, improving hardware utilization and reducing infrastructure costs.

Phase 2 – Storage Virtualization

Storage resources from multiple physical storage devices are combined into a centralized storage pool using Software Defined Storage (SDS). This allows dynamic allocation of storage, easier backups, and higher availability.

Phase 3 – Network Virtualization

Software Defined Networking (SDN) is introduced to separate network control from physical hardware. Virtual switches, routers, and firewalls are deployed, allowing network policies and configurations to be managed centrally.

Final Phase – Automation and Monitoring

Automation tools manage VM provisioning, scaling, backups, monitoring, and disaster recovery. Resource allocation becomes dynamic, improving scalability and reducing operational effort.

This phased approach minimizes downtime, lowers deployment risks, and enables gradual modernization of the datacenter.

4. Two cloud providers want to interoperate through federation. Explain how trust, identity, and privacy issues should be handled in the federation design.

Answer:

Cloud federation allows multiple cloud providers to share resources while maintaining secure communication.

Trust

A trusted relationship should be established between providers using digital certificates and Public Key Infrastructure (PKI). Service Level Agreements (SLAs) should clearly define responsibilities, security policies, and compliance requirements.

Management:

Identity

Federated Identity Management should be implemented using standards such as SAML, OAuth 2.0, or OpenID Connect. Users authenticate once and securely access services across both cloud providers through Single Sign-On (SSO).

Management:

Privacy

Sensitive data should be encrypted during transmission using TLS and while stored using strong encryption algorithms. Access should follow the Principle of Least Privilege with Role-Based Access Control (RBAC). Privacy regulations such as GDPR or HIPAA should be followed where applicable.

Protection:

Continuous auditing, logging, and monitoring should also be implemented to detect unauthorized access and ensure accountability across the federated environment.

5. An enterprise observes poor performance in a virtualized environment after overcommitting CPU and memory to multiple VMs. Analyze the scenario and recommend tuning actions.

Answer:

The enterprise has allocated more virtual CPUs and memory than the physical server can efficiently support. This overcommitment causes CPU contention, memory swapping, increased latency, and reduced application performance.

When many virtual machines compete for limited CPU resources, the hypervisor schedules them less efficiently, leading to slower response times. Memory overcommitment may force the hypervisor to swap memory to disk, significantly reducing performance.

The following tuning actions are recommended:

- Reduce CPU and memory overcommitment ratios.
- Allocate CPU and memory reservations for critical virtual machines.
- Increase physical RAM if resource demand is consistently high.
- Balance workloads using live migration across multiple hosts.
- Enable Dynamic Resource Scheduling (DRS) or load balancing.
- Monitor resource utilization continuously using virtualization management tools.
- Remove or shut down unused virtual machines to free resources.
- Optimize applications running inside virtual machines to reduce unnecessary resource consumption.

These tuning actions improve overall system performance, reduce resource contention, and ensure stable operation of the virtualized environment.